

Practical No. 2

Aim: To perform basic operations on Amazon EC2 including launching, monitoring, securing, accessing, resizing, and managing an EC2 instance using AWS.

Software Required: Amazon Web Services (AWS) Management Console

Theory:

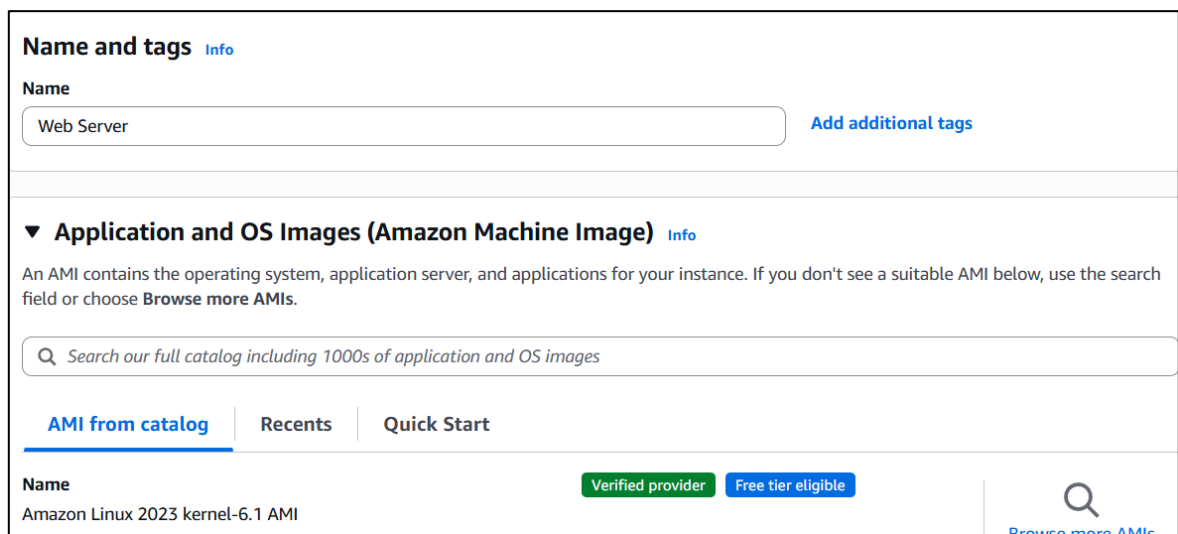
Amazon Web Services (AWS) is a cloud computing platform that provides on-demand access to computing resources such as servers, storage, databases, networking, and software through the internet.

Amazon Elastic Compute Cloud (EC2) is a core AWS service that provides resizable virtual servers known as instances. EC2 allows users to scale computing capacity up or down based on workload requirements.

Procedure:

Task 1: Launch Your Amazon EC2 Instance

- In the **AWS Management Console** choose **Services**, choose **Compute** and then choose **EC2** and select **Launch instance**.
- Under “*Name and tags*”, give the instance the name **Web Server**.
- For “*Application and OS images*”, keep the default **Amazon Linux** AMI selected.



The screenshot shows the AWS Management Console interface for launching an EC2 instance. The 'Name and tags' panel is at the top, with a text input field containing 'Web Server' and a link to 'Add additional tags'. Below this is the 'Application and OS Images (Amazon Machine Image)' panel, which includes a search bar and tabs for 'AMI from catalog', 'Recents', and 'Quick Start'. The 'AMI from catalog' tab is selected, showing a list of AMIs. The first AMI, 'Amazon Linux 2023 kernel-6.1 AMI', is highlighted, and it has 'Verified provider' and 'Free tier eligible' badges. A search icon and a link to 'Browse more AMIs' are visible at the bottom right of the AMI list.

- In the “*Instance type*” panel, keep the default **t2.micro** selected, and for For “*Key pair name*”, choose **vockey**.
- Next to “*Network settings*” choose **Edit**, and for “*VPC*” select **Lab VPC**.
- Under **Firewall (security groups)**, choose **Create security group** and configure:
 - Security group name:** Web Server security group
 - Description:** Security group for my web server
 - Under “*Inbound security group rules*”, click **Remove**.

▼ Network settings [Info](#)

VPC - required [Info](#)

vpc-04daf7443a69ba6cb (Lab VPC)
10.0.0.0/16

Subnet [Info](#)

subnet-0ea0093f4a3689c39 **PublicSubnet1**
VPC: vpc-04daf7443a69ba6cb Owner: 338633164042 Availability Zone: us-east-1a (use1-az1)
Zone type: Availability Zone IP addresses available: 1 CIDR: 10.0.1.0/28

Auto-assign public IP [Info](#)

Enable

Additional charges apply when outside of free tier allowance

Firewall (security groups) [Info](#)

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic

☒ Create security group ☐ Select existing security group

Security group name - required

Web Server security group

This security group will be added to all network interfaces. The name can't be edited after the security group is created. Allowed characters: lowercase letters, uppercase letters, numbers, hyphens, underscores, and equals signs. No spaces or special characters.

Description - required [Info](#)

Security group for my web server

Inbound Security Group Rules

No security group rules are currently included in this template. Add a new rule to include it in the launch template.

[Add security group rule](#)

g. In the “Configure storage” keep the default settings, and under “Advanced details” select **Enable** for “Termination Protection”.

h. Copy and paste the code below into the “User data” box:

```
#!/bin/bash
dnf install -y httpd
systemctl enable httpd
systemctl start httpd
echo '<html><h1>Hello From Your Web Server!</h1></html>' >
/var/www/html/index.html
```

User data - optional [Info](#)

Upload a file with your user data or enter it in the field.

[Choose file](#)

```
#!/bin/bash
dnf install -y httpd
systemctl enable httpd
systemctl start httpd
echo '<html><h1>Hello From Your Web Server!</h1></html>' > /var/www/html/index.html
```

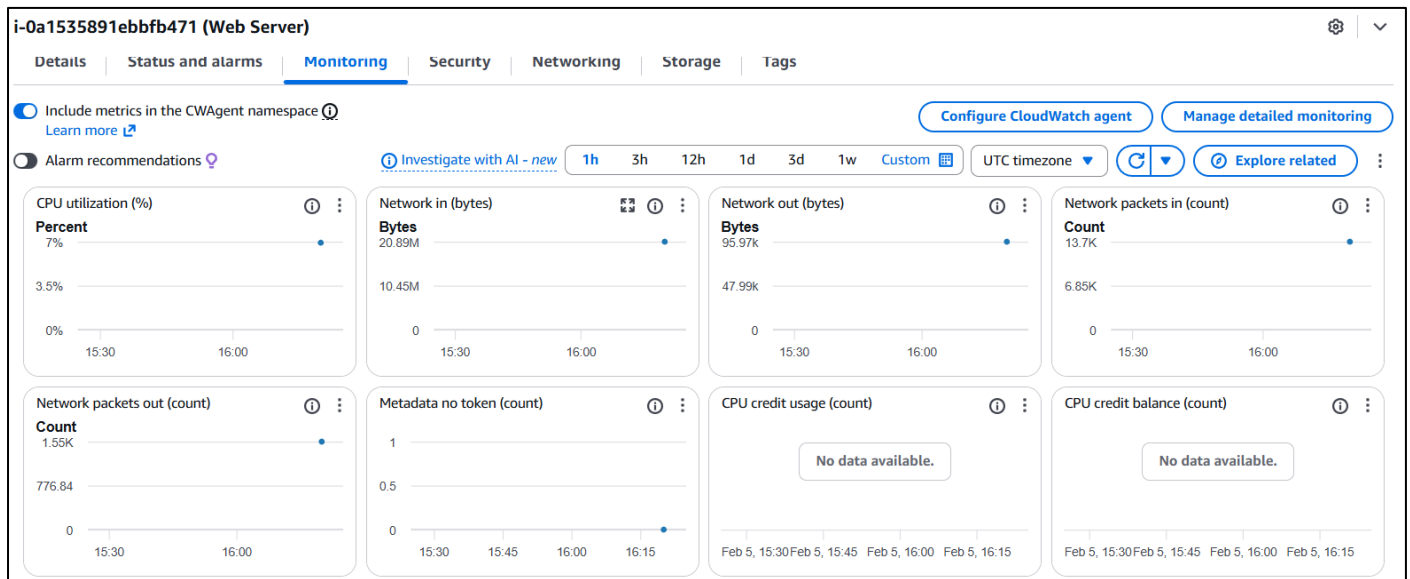
i. At the bottom of the “Summary” panel choose **Launch instance**, and wait for instance to display the following:

- **Instance State:** *Running*
- **Status Checks:** *2/2 checks passed*

	Name	Instance ID	Instance state	Instance type	Status check
☐	Bastion Host	i-01c3bd5a12800572d	Running	t2.micro	2/2 checks passed
☒	Web Server	i-0a1535891ebbf471	Running	t2.micro	2/2 checks passed

Task 2: Monitor Your Instance

- Select Web Server, and choose the **Monitoring** tab.



- In the “Actions” menu select **Monitor and troubleshoot Get system log**, and note that the HTTP package was installed from the “user data”.

System log Logs are only retrieved after an instance lifecycle event (launch/reboot/terminate/hibernate) [Copy log](#)

When you experience issues with your EC2 instance, reviewing system logs can help you pinpoint the cause.

```
[ 35.358004] cloud-init[2179]: mod_lua-2.4.66-1.amzn2023.0.1.x86_64
[ 35.358029] cloud-init[2179]: Complete!
[ 35.490956] cloud-init[2179]: Created symlink /etc/systemd/system/multi-user.target.wants/httpd.service → /usr/lib/systemd/system/httpd.service.
[ 35.190537] zram_generator::config[3658]: zram0: system has too much memory (961MB), limit is 800MB, ignoring.
ci-info: +-----+Authorized keys from /home/ec2-user/.ssh/authorized_keys for user ec2-user+-----+
ci-info: | Keytype | Fingerprint (sha256) | Options | Comment |
ci-info: +-----+-----+-----+-----+-----+
ci-info: | ssh-rsa | 48:fa:ab:88:09:34:56:dd:de:39:63:a4:64:49:a0:40:96:63:6b:a4:c4:3c:56:1e:d6:1b:33:49:75:22:5b:ba | - | vockey |
ci-info: +-----+-----+-----+-----+-----+
<14>Feb 5 16:22:04 cloud-init: #####
<14>Feb 5 16:22:04 cloud-init: -----BEGIN SSH HOST KEY FINGERPRINTS-----
<14>Feb 5 16:22:04 cloud-init: 256 SHA256:Tae6Yct0snUfjUy3TQRkQ7V92biIM8+QeTWktp94Dew root@ip-10-0-1-12.ec2.internal (ECDSA)
<14>Feb 5 16:22:04 cloud-init: 256 SHA256:sBFkyXw0HBRZuv4FexBC1Fcrk0RIM4e459vdowf3G8M root@ip-10-0-1-12.ec2.internal (ED25519)
<14>Feb 5 16:22:04 cloud-init: -----END SSH HOST KEY FINGERPRINTS-----
<14>Feb 5 16:22:04 cloud-init: #####
-----BEGIN SSH HOST KEY KEYS-----
ecdsa-sha2-nistp256 AAAAE2VjZHNhLXNoYTItbmlzdHh0YTYAAAAIbm1zdHh0YTYAAABBA7e1kjpMB16UKNe40Xef56WFC0jr/+O+SYEuwoV9tagmyZn+g7ZR/8DfRQ1ChlaofVZMF3BTtieqvRCmWMAQ= root@ip-10-0-1-12.ec2.internal
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAID91V2AZf/i/vrogUK2Fb8yv0T1Xa2egr+v4+1sRIYh root@ip-10-0-1-12.ec2.internal
-----END SSH HOST KEY KEYS-----
[ 36.123949] cloud-init[2179]: Cloud-init v. 22.2.2 finished at Thu, 05 Feb 2026 16:22:04 +0000. Datasource DataSourceEc2. Up 36.11 seconds
```

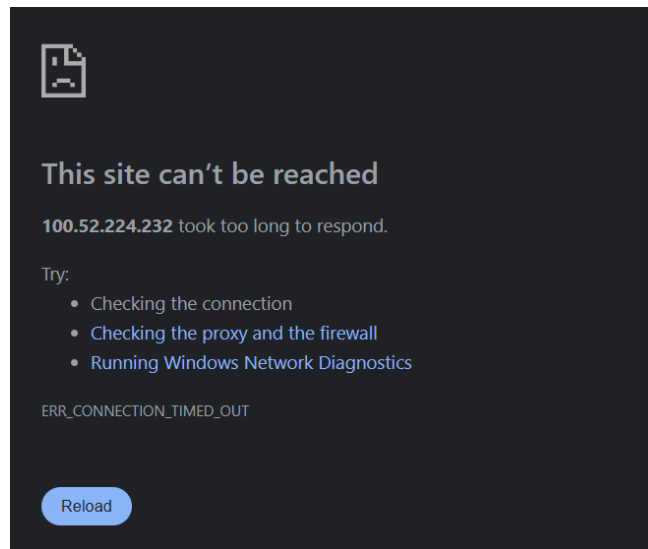
- Again in the “Actions” menu, select **Monitor and troubleshoot Get instance screenshot**.

Amazon Linux 2023.10.20260120
Kernel 6.1.159-182.297.amzn2023.x86_64 on an x86_64 (-)

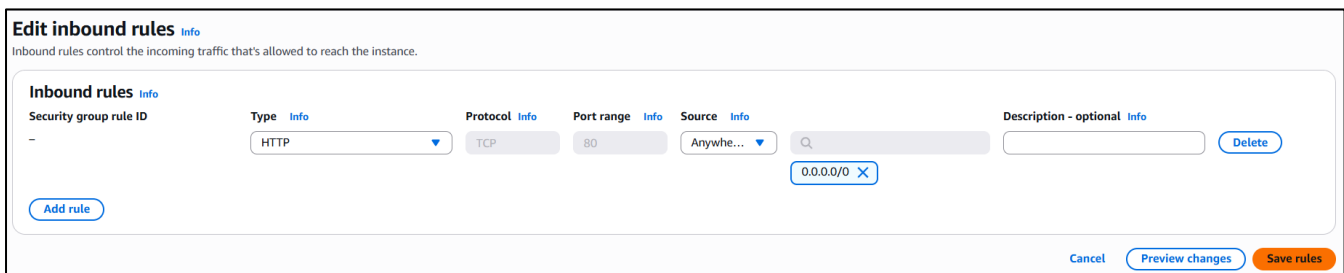
```
ip-10-0-1-12 login: [ 34.124188] zram_generator::config[2359]: zram0: system has too much memory (961MB), limit is 800MB, ignoring.
[ 35.190537] zram_generator::config[3658]: zram0: system has too much memory (961MB), limit is 800MB, ignoring.
```

Task 3: Update Your Security Group and Access the Web Server

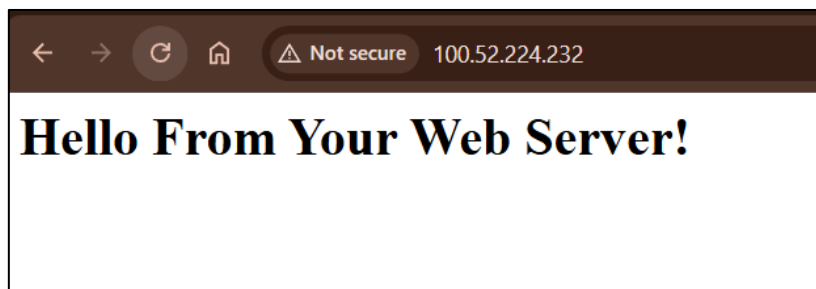
- Select “Web Server”, and choose the “Details” tab and Copy the **Public IPv4 address**.
- Open a new tab in a web browser, paste the IP address and press **Enter**. It will show you an error, so keep the browser tab open and return to the “EC2 Console” tab.



- c. Under “Security Groups” select “Web Server security group”, and choose the “Inbound rules” tab, choose “Edit inbound rules” and select **Add rule**:
- **Type:** HTTP
 - **Source:** Anywhere-IPv4
 - Choose **Save rules**

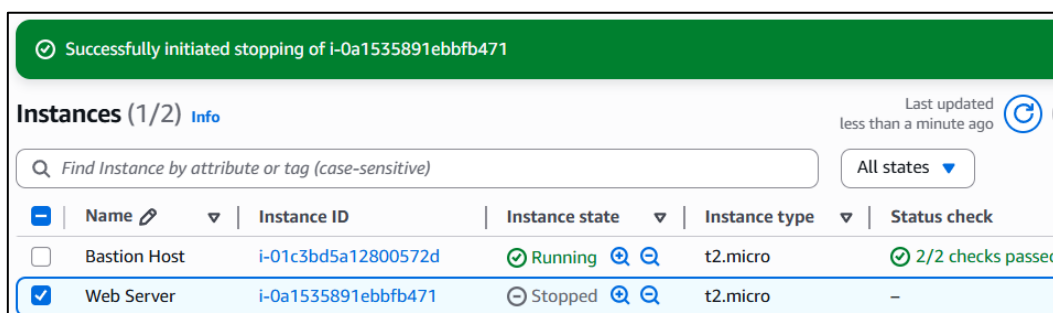


- d. Return to the web server tab that you previously opened and refresh the page. You should see the message “Hello from Your Web Server!”



Task 4: Resize Your Instance: Instance Type and EBS Volume

- a. Select the **Web Server** instance and in the “Instance state” menu, select **Stop instance** and wait for the **Instance state** to display: *Stopped*.



b. In the “Actions” menu, select “Instance settings Change instance type” then configure:

- **Instance Type:** *t2.small*
- Choose **Apply**

New instance type

Only instance types with x86 (32-bit) / (64-bit) architecture can be chosen.

c. Again in the “Actions” menu, select “Instance settings Change stop protection” and select **Enable**.

d. In the “Actions” menu, select “Modify volume”, the disk volume currently has a size of 8 GiB, change it to 10 GiB and choose **Modify**.

Volume details

Volume ID
 vol-0b617c28ab8a4eae7

Volume type | [Info](#)
General Purpose SSD (gp3)

Size (GiB) | [Info](#)

Min: 1 GiB, Max: 65536 GiB.

IOPS | [Info](#)

Min: 3000 IOPS, Max: 80000 IOPS.

Throughput (MiB/s) | [Info](#)

Min: 125 MiB, Max: 2000 MiB. Baseline: 125 MiB/s.

e. In left navigation panel choose “Instances” select the **Web Server** instance and in the “Instance state” menu select **Start instance**.

✔ Successfully initiated starting of i-0a1535891ebbf471

Instances (1/2) [Info](#) Last updated less than a minute ago

All states ▾

☐	Name ↗	Instance ID	Instance state	Instance type	Status check
☐	Bastion Host	i-01c3bd5a12800572d	✔ Running 🔍 🔍	t2.micro	✔ 2/2 checks passed
☒	Web Server	i-0a1535891ebbf471	✔ Running 🔍 🔍	t2.micro	🕒 Initializing

Task 5: Explore EC2 Limits

- In the “AWS Management Console”, in the search box next to “Services”, search for and choose **Service Quotas** and choose **AWS services**, and in “Find services” search bar, search for ec2 and choose **Amazon Elastic Compute Cloud (Amazon EC2)**.
- In the *Find quotas* search bar, search for running on-demand, but do not make a selection.

Service quotas

info

Request increase at account level

View your applied quota values, default quota values, and request quota increases for quotas. [Learn more](#)

Q

running on-demand

X

10 matches

<

1

>

⚙

	Quota name	Applied account-level quota value	AWS default quota value	Utilization	Adjustability
☐	Running On-Demand DL instances	96	0	0	Account level
☐	Running On-Demand F instances	64	0	0	Account level
☐	Running On-Demand G and VT instances	0	0	0	Account level
☐	Running On-Demand High Memory instances	0	0	0	Account level
☐	Running On-Demand HPC instances	192	0	0	Account level
☐	Running On-Demand Inf instances	8	0	0	Account level
☐	Running On-Demand P instances	0	0	0	Account level
☐	Running On-Demand Standard (A, C, D, H, I, M, R, T, Z) instances	256	5	1	Account level
☐	Running On-Demand Trn instances	8	0	0	Account level
☐	Running On-Demand X instances	0	0	0	Account level

Task 6: Test Stop Protection

- In left navigation pane choose “*Instances*” and select the **Web Server** instance and in the “*Instance state*” menu select **Stop instance**, it will show an error.

❌ Failed to stop the instance i-0a1535891ebbf471
The instance 'i-0a1535891ebbf471' may not be stopped. Modify its 'disableApiStop' instance attribute and try again.

Instances (1/2) Info

All states ▼

☐	Name	Instance ID	Instance state	Instance type	Status check
☐	Bastion Host	i-01c3bd5a12800572d	✔ Running	t2.micro	✔ 2/2 checks passed
☒	Web Server	i-0a1535891ebbf471	✔ Running	t2.micro	✔ 2/2 checks passed

- b. In the “*Actions*” menu select “*Instance settings Change stop protection*” and remove the check next to **Enable**.
- c. Select the **Web Server** instance again and in the “*Instance state*” menu, select **Stop instance**.

🔄 Successfully initiated stopping of i-0a1535891ebbf471

Instances (1/2) [Info](#)

🔍 Find Instance by attribute or tag (case-sensitive)

🔄 Last updated less than a minute ago

🔼 All states 🔽

⊞	Name ✎	Instance ID	Instance state 🔽	Instance type 🔽	Status check
☐	Bastion Host	i-01c3bd5a12800572d	✅ Running 🔍 🔍	t2.micro	✅ 2/2 checks passed
☒	Web Server	i-0a1535891ebbf471	⏸ Stopped 🔍 🔍	t2.micro	–

Conclusion: Hence, we have successfully performed basic operations on Amazon EC2 including launching, monitoring, securing, accessing, resizing, and managing an EC2 instance using AWS.